Retrieve decimal and percentage methods

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: __________________

1. Read the model: Find 37.5% of 640. Copy or complete the first step.

2. Simplify 45:60 and find an equivalent ratio with first part 12.

3. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find 37.5% of 640. Copy or complete the first step.
   Answer 240.

2. Simplify 45:60 and find an equivalent ratio with first part 12.
   Answer 3:4 and 12:16.

3. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer 1 3/4 kg.

4. Miro says: Miro chooses a familiar method before identifying the number type and relationship. Is this correct? Explain.
   Answer Name the domain first, represent the relationship, then choose and check the calculation.

Retrieve decimal and percentage methods

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Find 37.5% of 640.

2. Simplify 45:60 and find an equivalent ratio with first part 12.

3. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

4. Solve this independently and show every stage: Find 37.5% of 640.

5. Use a second method or representation for: Simplify 45:60 and find an equivalent ratio with first part 12.

6. Check this result using an inverse, estimate or known fact: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

7. Prove or explain your reasoning: Order 0.72, 7/10 and 73%.

8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

Teacher answers and checking prompts

1. Find 37.5% of 640.
   Answer 240.

2. Simplify 45:60 and find an equivalent ratio with first part 12.
   Answer 3:4 and 12:16.

3. A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer 1 3/4 kg.

4. Solve this independently and show every stage: Find 37.5% of 640.
   Answer 240.

5. Use a second method or representation for: Simplify 45:60 and find an equivalent ratio with first part 12.
   Answer Expected result: 3:4 and 12:16. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer Expected result: 1 3/4 kg. A valid check must be shown.

7. Prove or explain your reasoning: Order 0.72, 7/10 and 73%.
   Answer 7/10, 0.72, 73%. The explanation must show why the method works.

8. Identify and correct this misconception: Miro chooses a familiar method before identifying the number type and relationship.
   Answer Name the domain first, represent the relationship, then choose and check the calculation.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Retrieve decimal and percentage methods

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ______________________________ Date: ____________

1. Prove or explain your reasoning: Order 0.72, 7/10 and 73%.

2. Prove the answer to this question, rather than only calculating it: Find 37.5% of 640.

3. Solve this in two different ways and compare them: Simplify 45:60 and find an equivalent ratio with first part 12.

4. Work backwards from the answer to reconstruct the question: 1 3/4 kg.

5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.

6. Create a new example that tests the same idea as: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Order 0.72, 7/10 and 73%.
   Answer 7/10, 0.72, 73%. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: Find 37.5% of 640.
   Answer Expected result: 240. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Simplify 45:60 and find an equivalent ratio with first part 12.
   Answer Expected result: 3:4 and 12:16. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 1 3/4 kg.
   Answer One valid reconstruction is: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people? Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro chooses a familiar method before identifying the number type and relationship.
   Answer Name the domain first, represent the relationship, then choose and check the calculation.

6. Create a new example that tests the same idea as: A recipe uses 3/4 kg for 6 people. How much is needed for 14 people?
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.